

REFERENCE DOCUMENT

Conceptual Architecture & Tech Stack Questions

An interview-style reference covering stack selection, backend runtimes, data stores, distributed systems, delivery and AI architecture. Each question is paired with an answer written for defensible decision-making rather than recall.

Format	Question and answer
Sections	18
Questions	313
Difficulty	Junior, Mid, Senior / Architect — marked per question
Companion to	Vibe Coding Pre-Build Template (TRD, Backend Schema)

CONTENTS

01	Tech Stack Selection Fundamentals	1–20
02	Node.js vs FastAPI vs Others (Backend Runtime)	21–48
03	Frontend Framework Trade-offs	49–70
04	Rendering Strategy: SSR / SSG / ISR / CSR / Edge	71–85
05	Database Selection & Trade-offs	86–115
06	ORM & Data Access Layer	116–127
07	Architecture Patterns: Monolith / Microservices / Serverless	128–152
08	API Design: REST / GraphQL / gRPC / tRPC / WebSockets	153–172
09	Load Balancing	173–205
10	Caching & CDN	206–220
11	Scaling, Queues & Async Processing	221–233
12	Auth & Security Architecture	234–243
13	Deployment, Observability & Cost Architecture	244–250
14	Networking & Protocol Fundamentals	251–265
15	File Storage & Media Architecture	266–277
16	Payments, Billing & Third-Party Integrations	278–289
17	AI / LLM Application Architecture	290–306
18	Environments, Config, Feature Flags & CI/CD	307–313